Storage

Introduction

In everyday computer use, we need a way to write data, rewrite data, store data when the computer is switched off and retrieve data the next time we switch our computer back on. We have already looked at system/primary memory:

- RAM allows us to write data and store it...but the data is lost when it is switched off (volatile).
- ROM allows us to store data when it has no power (non-volatile)...but we cannot write to it.

We therefore need another device/medium by which we can re-write data when we want AND store the data when the computer has no power. This is where SECONDARY STORAGE comes in!

Storage Technologies:

Magnetic storage



Magnetic storage uses different patterns of magnetisation in a magnetisable material to store data (tape/hard disk). Magnetised points on a pattern represents '1' and demagnetised points represent '0'

Optical storage

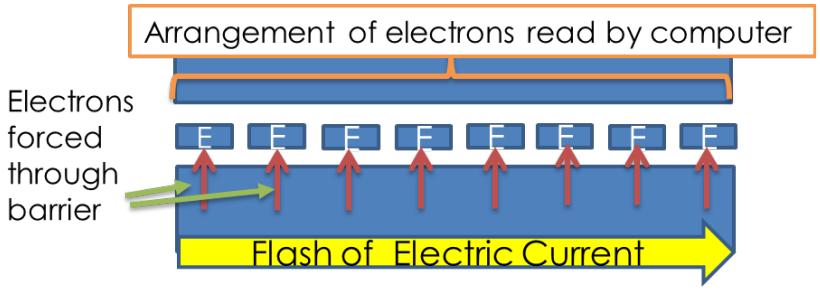


Optical storage consists of 'marks' arranged in patterns. When writing to optical media the laser burns 'pits' into the shiny surface (land). When reading optical media a light (laser) is shined on its surface. When it hits land the beam reflects (0) but when it hits the pits, it doesn't (1).

Solid State Storage (Flash)



They work by sending a large electric current to force electrons through a barrier and trapping them in position. The flash of the electric current used to achieve this gives us the name for this type of memory. The different arrangement of electrons gives us data.



Magnetic Storage Performance

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Hard Disk
(Hard Drive)

Use	Desktop computers to store large files, programs and operating systems
Capacity	750GB – 3TB (3000GB) (Large)
Speed	High read/write speeds
Portability	Heavy / poor portability (even external are clunky!)
Durability	Dropping this device could cause damage, especially to the 'read head'
Reliability	Can be used again and again but does have a limited life (a good few years)
Cost	Low cost (quite cheap)



Magnetic
Tape

Use	Backing up vast amount of data
Capacity	1 – 5TB (Very Large)
Speed	High write speeds BUT Slow read speeds
Portability	Fairly small and light (but reading device is not because if the data you need is at the end of the tape you have to read/play the tape from the start – which takes time (known as serial access))
Durability	Fairly durable.
Reliability	Can be used again and again.
Cost	Cheap

Optical Storage



CD-ROM

Use	Storage of small media files and documents
Capacity	650MB (Small)
Speed	Moderate read/write speeds
Portability	Very portable and light
Durability	Fair durability, can survive the odd knock but scratches can damage the data.
Reliability	CD-Rs are write once but can be read over and over. (CD-RW can be reused)
Cost	Very cheap

BLU-RAY



DVD-ROM

Use	Storage of larger media files and documents
Capacity	4.37GB (DVD) 25GB (Standard Blu-Ray) (Small)
Speed	Moderate read/write speeds
Portability	Very portable and light
Durability	Fair durability, can survive the odd knock but scratches can damage the data.
Reliability	DVD-Rs are write once but can be read over and over.
Cost	Very cheap

Solid State / Flash Storage



Solid State Drive

Use	Desktop computers, laptops and tablets to store large files, programs and operating systems
Capacity	64GB – 480GB (Moderate)
Speed	Super Fast (Read/Write)
Portability	Very portable (small and lightweight)
Durability	No moving parts so can survive knocks and scrapes
Reliability	Can be re-used many times (up to 100,000 times)
Cost	Very Expensive (per Gb)



Flash
SD Card

Use	ones, Cameras (portable devices) – great oice as no moving parts so mobile devices can ction whilst on the move
Capacity	16GB – 64GB
Speed	Super Fast (Read/Write)
Portability	Extremely portable (small and lightweight)
Durability	No moving parts so can survive knocks and scrapes
Reliability	Can be re-used many times (up to 100,000 times)
Cost	Expensive (per Gb)

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Flash Memory Stick

Use	Backing up or transferring documents, small media files from computer to computer
Capacity	1GB – 64GB
Speed	Super Fast (Read/Write)
Portability	Extremely portable (small and lightweight)
Durability	No moving parts so can survive knocks and scrapes
Reliability	Can be re-used many times (up to 100,000 times)
Cost	Expensive (per Gb)

Network Attached Storage (NAS)



NAS Drive

Use	A stand alone external hard drive at the centre of a network. Plugs into the network to allow users of different computers to write or read files. Usually documents and media files.
Capacity	750GB – 3TB (3000GB) (Large)
Speed	Hard disk has high read/write speeds The network speed (transfer of data from computer to NAS) can cause slow data access speeds
Portability	Heavy / poor portability.
Durability	Dropping this device could cause damage, especially to the 'read head'
Reliability	Can be used again and again but does have a limited life (a good few years)
Cost	If Hard Disk is used then storage is cheap If SS Disk is used then storage is expensive In addition to the storage, the NAS enclosure can be very expensive depending on its features.

Cloud Storage (online storage)

Use	Used to store documents and media files online. This means that you can go to another computer and access your files (providing you have internet access).
Capacity	Unlimited (providing you can pay for it).
Speed	Depends on network / bandwidth speeds Poor bandwidth can cause uploading and downloading of large files to take a long time
Portability	It is virtual so you don't have anything physical to move around.
Durability **Reliability**	Providing you keep up payments and the company looks after their storage devices – very durable and reliable. This is often because they back up storage devices in many places. There can be issues with the law as a company's data may be stored in a country which doesn't follow the same data protection laws. The hosting company could also me attached by hackers who could steal your data.
Cost	Can be expensive depending on the service – but often it is free for a few Gbs.









Questions *(The question zone you choose must either match your target grade or be higher!)*

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Question Zone 1-3

1	2	3	4	5	6	7	8	9

1. State the 3 different storage technologies. [3]
2. Describe how optical storage devices store data. [3]
3. What is the need for secondary storage if a computer system already has primary memory (RAM/ROM)? [3]

Question Zone 4-6

1	2	3	4	5	6	7	8	9

1. What is the need for secondary storage if a computer system already has primary memory (RAM/ROM)? [3]
2. Which secondary storage technology would be most appropriate for a mobile phone? Explain your answer. [4]
3. Magnetic tape storage can store large capacities but there is a major drawback when it comes to reading data from this technology. Explain the drawback. [3]

Question Zone 7-9

1	2	3	4	5	6	7	8	9

1. Magnetic tape storage can store large capacities but there is a major drawback when it comes to reading data from this technology. Explain the drawback. [3]
2. Explain which storage device is best suited to storing an operating system. [5]
3. Discuss the relative advantages and disadvantages of the 3 main storage technologies (optical, magnetic (hard disks), solid state). [5]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Magnetic Storage Technology: *Using materials with magnetic properties (ability to change polarity (north/south)) to store data (0/1).*

Optical Storage Technology: Using reflective material to store data by burning 'pits' and leaving 'land'. Data is read by shining a laser and seeing which elements reflect (0) and which do not (1).

Solid State Storage Technology*: Using circuitry, which traps electrons into specific orders in order to store data.*

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Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
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Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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